Student housing near VUB - investment decision

Valuation: 29 October 2026. All figures nominal, pre-tax. Hypothetical site; no investment commitment.

Base project NPV: EUR -4,384,562. Project IRR: 4.95%.

Best constructed case NPV: EUR -2,769,706. Selected NPV including no investment: EUR 0.

Do not invest: zero NPV dominates

Best built strategy index 1 (1=A,2=B,3=C); gates for DSCR and sustainability still require approval. No-build avoids negative investment NPV.

Evidence and assumptions

VUB standard 2026-27 institutional room: EUR 449.30/month including services. Brukot city ordinary-room median: EUR 590/month including charges. Private residence rents can be higher. These do not establish achieved rent for this site.

Brussels property-specific reference rent and planning review remain outstanding. The 20% excessive-rent presumption is qualified, not a universal price ceiling.

Construction EUR 2200/m2, demand 125 students, elasticity 1.2, land EUR 1.5m and 4.5% borrowing are editable management assumptions, not observed site data.

Business model and reliability

Demand = 125 * demand shock * exp[-1.2*(rent/590 - 1)]. Occupied market rooms are capped at available rooms. Affordable rooms have separate prices and demand.

One period of development at year 0, twenty annual operating periods. NOI, replacement capex, debt amortization and terminal proceeds are separate. No corporate income taxes or tax shields.

Project discount rate 8%, equity hurdle 10%, common nominal growth 2%, exit cap 5.5% are unverified investor assumptions.

Managerial decisions - optimization and break-even

Capacity and rent: systematic integer search finds best built 110 rooms, EUR 650/month market rent. No-build remains available.

Maximum viable development spend at base operating economics: EUR 9,599,877. This is a ceiling, not a minimum required investment.

Undiscounted payback including sale: 19.070634212891168; discounted payback: Beyond horizon.

Optimization uses capacities 40-140, EUR 10 rent steps, floor area and budget limits, and 10% affordable rooms. A continuous global optimum and plot-specific feasibility are not claimed.

Sensitivity and uncertainty

Largest NPV swing over stated input envelopes: Hard cost EUR/m2. Comparing different ranges does not establish statistical causality.

5000 independent persistent triangular trials: mean NPV EUR -5,110,182, P5 EUR -7,293,373, P95 EUR -3,106,171.

Probability of negative NPV: 100.0%. This probability is conditional on subjective distributions, not a forecast calibrated to market history.

Cost factor 0.85/1/1.25; demand 0.75/1/1.15; opex 0.9/1/1.2; exit cap 4.5/5.5/7%. Triangular distributions reflect judgmental min/mode/max. Correlated adverse shocks and delays could worsen tails.

Fifth simultaneous uncertainty: common annual rent/opex growth, triangular 0.5%/2%/3.5%. Mean project IRR 4.45%; undefined IRRs 0; fraction meeting 8% hurdle 0.0%. Full static trial IRRs in CSV appendix; rerun after changes.

Affordability, sustainability and recommendation

Affordable share 10.0%. Affordable rent EUR 449.30 including services. At least 10% is an explicit management constraint; eligibility assumed means-tested enrolled students.

Illustrative annual operational carbon: 6.5 tonnes; energy 216,000 kWh. Energy and emissions factors are not measured or certified.

Obtain a secured site, itemized construction bids, achievable rent survey, pre-letting evidence, planning/fire/EPC review and lender terms before approval. Reject projects with nonpositive NPV or failed constraints.

Do not force profitability. Student review of financial assumptions, sources and formula logic remains required.

A/B/C modelled energy = 2400/1800/1100 kWh per room; construction premiums 0%/6%/12%; affordable shares 10%/10%/20%. Illustrative engineering alternatives, not certifications.

References and implementation limits

https://www.vub.be/en/studying-vub/practical-info-for-students/student-housing/practical-housing-information

https://www.brukot.be/en/student-guide/rent-stats

https://be.brussels/fr/logement/location/bail-dhabitation/loyers-de-reference

https://abex.be/en/abex-index/?v=910955a907e7

https://www.nbb.be/en/statistics/topics/financial-institutions/banks/interest-rates-loans-and-deposits

Formula-based sensitivity and finite grid optimization implemented; native Excel Data Tables and Solver not configured. Cached values supplied by a local supported-function evaluator. Microsoft Excel/Word/PowerPoint desktop execution not performed. Report and slides are concise text-based documents; charts are in the workbook.

Full requested rubric is not complete: financing choice, delays, native tables/Solver and empirical calibration remain. Source benchmarks are verified; group review and exact tax/legal conclusions are outstanding.